

ASA Seminar

Existence of solutions for the nonlocal problem involving the nonlocal $(N, N/s)$ -Laplacian and nonlocal $p(x)$ -Laplacian with logarithmic and critical exponential nonlinearities

Speaker

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Recently, by applying variational methods, Y. Yang et al. obtained the existence of solutions for the following (p, N) -Laplacian equation with logarithmic and critical exponential nonlinearities

$$-\Delta_p u - \Delta_N u = |u|^{q-2} u \ln |u|^2 + \lambda f(u) \text{ in } \Omega, \quad u = 0 \text{ on } \partial\Omega \quad (0.1)$$

where $\Omega \subset \mathbb{R}^N$ is a bounded domain, $N \geq 2$, $1 < p < N < q$, $\lambda > 0$ is a positive real parameter.

The main objective of this presentation is to prove the existence of a solution of equation (0.1) for the case (p, N) -Laplacian is now replaced by a non-linear combination of fractional $p(x)$ -Laplacian, fractional N -Laplacian and fractional $\frac{N}{s}$ -Laplacian operators, in particular the operator $\mathcal{L}_{p,N}^s$ which we define. Our approach combines variational methods based on critical point theory. It is essential to rely on relevant findings made for instance by Azroul, Bahrou and DelPezzo regarding fractional Sobolev spaces with different exponents, and the fractional $p(x)$ -Laplacian operator.

For this purpose, we consider the problem

$$\mathcal{L}_{p,N}^s u = g(u) \text{ in } \Omega, \quad u = 0 \text{ in } \Omega^c, \quad (0.2)$$

where $g(u) := |u|^{q(x)-2} u \ln |u|^2 + \lambda f(u)$ with $q(x) > \frac{N}{s}$ for every $x \in \overline{\Omega}$. We start by fixing $s \in (0, 1)$ and let $p : \overline{\Omega} \times \overline{\Omega} \rightarrow (1, +\infty)$ be a continuous bounded function such that $p_+ < \frac{N}{s}$ and $\lambda > 0$ is a positive real parameter. Moreover, we assume that

$f : \mathbb{R} \rightarrow \mathbb{R}$ is a function in C^1 with exponential growth behaving like $e^{\alpha|z|^{\frac{N}{N-s}}}$ as $z \rightarrow \infty$ for some $\alpha > 0$.